Reasoning with Generative AI (Fall 2026, DATA 101)

Schedule
Section 3: TR 9:30am - 10:50am
Section 4: TR 11:00am - 12:20pm

Instructor
Dr. Rohaifa Khaldi
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ISC4 3313

Office Hours:
Instructor: Friday 11am - 1pm or by appointment.
GTA: TBD (located in the SSC).

Prerequisite(s):
N/A

Credit Hours: 3

Course Description

This course introduces students to computational thinking, data reasoning, and generative AI through hands-on programming and creative problem solving. Students learn how computers represent and process information using Python and modern data tools while exploring how generative AI systems can support coding, automation, and innovation. Topics include programming fundamentals, numerical computing, data processing, and human-AI collaboration.
Throughout the course, students will engage in live coding activities, AI-assisted exercises, and project-based learning experiences designed to develop computational reasoning, creativity, and critical thinking skills. By the end of the course, students will design and build original AI-assisted applications that demonstrate responsible and effective use of genAI technologies.

Course Outline

The course outline is provided below. Weekly content may be adjusted as needed to reflect the pace and progress of the class.

Table 1: Course Modules and Approximate Schedule

Module	**Topic**	**Duration**
Module 1	Computational Thinking with Python	$\sim$ 4 weeks
Module 2	Numerical Data and Computational Representation	$\sim$ 3 weeks
Module 3	Tabular Data and Data Processing	$\sim$ 5 weeks
Final Project		

Expectations

All students are strongly encouraged to bring their laptops/notebook computers to class since we plan on having a significant amount of active coding.
Students are expected to attend all lectures regularly and actively engage with the course material. They should ensure that all activities are delivered on time. In addition, students are encouraged to participate in discussions, collaborate respectfully with peers, and take responsibility for keeping up with readings and practice exercises outside of class. Consistent preparation and active involvement will be essential for your success in this course.

Attendance

Attendance is encouraged for this course, as class sessions will include live coding activities, hands-on exercises, and discussions that are designed to support student learning and project development. Regular participation will help students stay engaged with the course material and successfully complete course assessments.

Classroom Behavior

Students are expected to engage respectfully during discussions to support the open exchange of ideas. All perspectives are welcome, and students should practice active, attentive listening to maintain a supportive learning environment. Disruptive behaviors, including loud conversations with peers or other actions that interfere with class activities, are not permitted. Eating during class sessions is not allowed in order to help maintain a clean, focused, and respectful classroom environment.

Blackboard Course Management System

Blackboard will be used as the primary learning management system for this course. Course materials, including lecture slides, challenges, quizzes, and grades will be posted on Blackboard.

Communication Platforms

Course communication will take place through the following platforms:

- **Blackboard Announcements:** Important updates, reminders, and schedule changes will be posted through Blackboard announcements.
- **Module Agendas:** Each module includes an agenda outlining the planned lectures, activities, assessments, and deadlines. Students are responsible for checking module agendas regularly, as they may be updated due to academic, environmental, technological, or other circumstances.
- **Module Discussions:** Each module includes a Questions and Discussion forum where students may ask questions about course content, quizzes, challenges, and projects. Students should review existing posts before submitting a new question.
- **Email:** Email should be used for individual or confidential matters, including grades, absences, accommodations, extensions, or other personal concerns.
- **Office Hours:** Students may attend office hours for questions requiring detailed explanations or individualized assistance.

Students are responsible for regularly checking Blackboard, course announcements, and module agendas. They should not post assessment answers, challenge solutions, personal information, or grade-related questions in module discussions. Only information provided by the instructor or GTA should be considered official course guidance. Please allow up to two business days for a response.

Communication Policy

For all questions or concerns related to quizzes or challenges, students should first communicate with the Graduate Teaching Assistant (GTA). The GTA is the primary point of contact for course-related issues and is equipped to address most questions. If an issue cannot be resolved at that level, the GTA will escalate the matter to the instructor as appropriate. Adhering to this communication process helps ensure fairness, consistency, and efficient course administration. Confidential or sensitive matters may be communicated directly to the instructor.

When sending emails to the course staff, students must include the course number (DATA 101) and their section number (Section 3 or 4) in the **email subject line**. This helps ensure that messages are properly routed and addressed in a timely manner.

Please give 48 hours (usually 24 hours or less) for faculty to respond to email on weekdays. Emails sent on the weekend will be responded to on the following Monday.

Student Accessibility Services (SAS)

William & Mary accommodates students with disabilities in accordance with federal laws and university policy. Any student needing accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services at 757.221.2512 or at sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation.

Student Success Center (SSC)

Beginning next week, the Student Success Center (SSC) will be available to all Data Science students on weekdays from 10:00 a.m. to 8:00 p.m. It is located on the third floor of ISC 4. After exiting the elevator, the center is immediately on the right.
Graduate and undergraduate tutors are available to:

- Assist with any Data Science course
- Answer general academic and orientation questions
- Support students in navigating the Data Science program

The SSC also provides a well-equipped and supportive environment for independent or collaborative study.
All students should check in before using the center, whether they are seeking assistance or using the space for self-study. To check in, scan the SSC QR code and complete the brief form, which takes approximately 30 seconds.

Mental & physical well-being

There are many resources to help navigate emotional, psychological, physical, medical, and material concerns, including:

- Counseling Center at (757) 221-3620. Services are free and confidential.
- Health Center at (757) 221-4386.
- The free Timely Care app gives students access to 24-7 remote counseling support.

Assessment Components

Quizzes
At the end of each module, students will complete a quiz designed to assess their understanding of the key concepts covered in that module. There will be a total of **three quizzes**.

Mini-Projects
Throughout the course, students will complete **three mini-projects**, one at the end of each module. These projects are designed to provide opportunities to apply the concepts and skills developed in each module by building practical, interactive applications.

Students are encouraged to use Gen AI tools as development partners to assist with tasks such as generating code, debugging, improving functionality, and developing application features. However, students remain fully responsible for the final work they submit. All AI-generated content and code must be carefully reviewed, tested, and validated for accuracy, correctness, functionality, and quality.

Students should be able to understand and explain their implementation and demonstrate how the concepts covered in the course are reflected in their application.

Final Project
At the end of the course, students will complete a final project that integrates and applies the knowledge and skills developed throughout the semester. Students will work in pairs to design and develop a meaningful, creative, and sufficiently challenging application using Python and Generative AI tools. The project should demonstrate computational thinking, effective use of course concepts, and the ability to use GenAI as a development partner to build functionality beyond what students could reasonably implement independently at this stage.
The final project will include the following components:

- **Project selection:** Each pair will select an application of their choice. The proposed application should address a meaningful need or problem and be sufficiently challenging to demonstrate the integration of concepts and skills developed throughout the course.
- **Application development:** Each pair will design and implement a functional application using Python and appropriate libraries or frameworks. Students are encouraged to explore advanced features and functionalities with the assistance of Generative AI tools.
- **Written report:** Each pair will submit a report describing the motivation and goal of the project, application design and functionality, implementation, use of Generative AI, challenges encountered, limitations, possible improvements, and conclusions. The report should also include screenshots demonstrating the main features of the final application.
- **Recorded presentation:** Each pair will submit a short recorded presentation introducing the application, demonstrating its main features, explaining important design and implementation decisions, and reflecting on the development process.

- **Use of GenAI:** Students are expected to use Generative AI tools to support the development of their application. GenAI may be used for tasks such as generating and improving code, debugging, designing application features, exploring alternative solutions, and supporting the development process. Students must disclose the tools used and briefly explain how they contributed to the project. Students remain fully responsible for understanding, testing, validating, and ensuring the accuracy, functionality, and quality of all AI-generated content and code.
- **Online oral defense:** After submitting the project, both students will attend a brief online meeting and answer questions about their work. Each student will be asked at least one question and may be asked to explain any part of the project, including the code, application functionality, design decisions, use of GenAI, challenges, limitations, or possible improvements.
- **Individual accountability:** Both partners must contribute meaningfully to the project and understand the entire application, regardless of how the work was divided. The project deliverables will receive a shared grade, while performance during the oral defense will be evaluated individually.
- **Evaluation:** Students will be evaluated on the quality and relevance of the project idea, application functionality, appropriate integration of course concepts, effective and responsible use of Generative AI, quality of the implementation, critical reflection on limitations and improvements, quality of the report and presentation, and their ability to explain and defend their work.

Grade Distribution

>= 93.00	A	73.00 - 76.99	C
90.00 - 92.99	A-	70.00 - 72.99	C-
87.00 - 89.99	B+	67.00 - 69.99	D+
83.00 - 86.99	B	63.00 - 66.99	D
80.00 - 82.99	B-	60.00 - 62.99	D-
77.00 - 79.99	C+	<= 59.99	F

Quizzes	25%
Mini Projects	30%
Final Project	45%

Quiz Policy

Quizzes will be administered during scheduled class sessions unless otherwise stated. Students who miss a quiz must complete it during the next available office hours and will receive a **10% penalty**. Failure to do so will result in a grade of zero. Exceptions will be considered only in documented extenuating circumstances.

Project Submission Policy

Projects are expected to be submitted by the stated deadlines. Projects will incur a **10% penalty per day for up to three days**. Submissions more than three days late will not be accepted and will receive a grade of zero. Exceptions will be considered only in documented extenuating circumstances.

Academic Integrity

All class-related activities are governed by the William & Mary Honor Code.

Use of GenAI

The use of generative AI (GenAI) is not permitted during quizzes unless explicitly stated otherwise by the instructor.
Students may use GenAI tools for challenges and the final project to support the development of more **advanced** workflows, analyses, and applications. However, students remain fully responsible for the accuracy, quality, and originality of their submitted work. They must understand the methods and code used, critically evaluate AI-generated content, and be able to explain and justify their decisions, implementation, and results.
Students may be asked to discuss or demonstrate their work and explain the process used to produce it. Submitting work that a student does not understand or cannot explain, or using GenAI where it is prohibited, may constitute an Honor Code violation and may be submitted for disciplinary review.

Important Dates

- Add/drop deadline: September 4.
- Labor Day (no classes): September 7.
- Fall Break: October 8–11.
- Withdrawal deadline: October 26.
- Election day (no classes): November 3.
- Remote Instruction Days (Courses must be delivered remotely, offices closed): November 23–24.
- Thanksgiving (no class): November 25-29.
- Last day of classes: December 4.
- Final Project deadline: TBD.